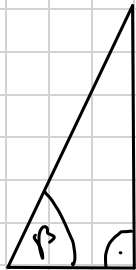
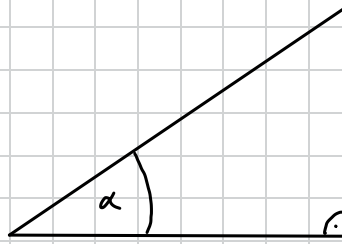
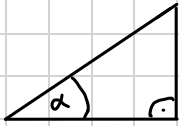


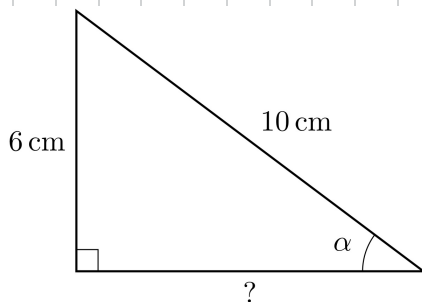
### exercise 1:

For each triangle, mark the hypotenuse with H, the side opposite the marked angle with O and the adjacent side A. Calculate  $\sin$  and  $\cos$  of the angles. What do you observe regarding similar triangles?



### exercise 2:

A right triangle has a hypotenuse of 10 cm. The side opposite the angle  $\alpha$  has length 6 cm.



a) use Pythagoras' theorem to calculate the adjacent side

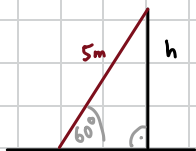
b) calculate  $\sin(\alpha)$  and  $\cos(\alpha)$

c) calculate  $\sin^2(\alpha) + \cos^2(\alpha)$   
Do you think the result is just a coincidence?

### exercise 3:

A ladder of length 5 m leans against a wall. The angle between ladder and the ground is  $60^\circ$ .

How can you calculate the height  $h$ ?  
Write down an equation for  $h$



exercise 4:

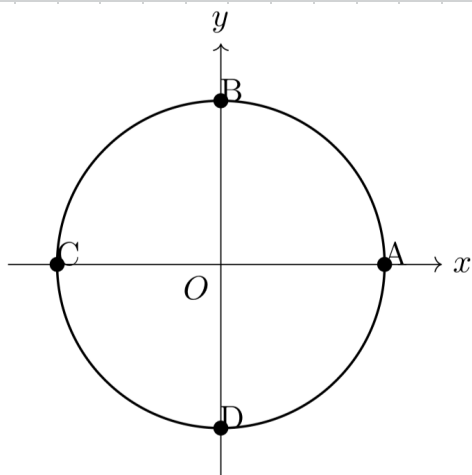
Complete the table. Remember that  $360^\circ = 2\pi$

| Degrees    | Radians | Degrees     | Radians          |
|------------|---------|-------------|------------------|
| $0^\circ$  |         | $180^\circ$ |                  |
| $30^\circ$ |         | $270^\circ$ |                  |
| $45^\circ$ |         | $360^\circ$ |                  |
| $60^\circ$ |         |             | $\frac{3\pi}{2}$ |
| $90^\circ$ |         |             | $\frac{\pi}{4}$  |

exercise 5:

The radius of the unit circle is 1. For an angle  $\alpha$ , the point on the circle has coordinates  $(\cos \alpha, \sin \alpha)$

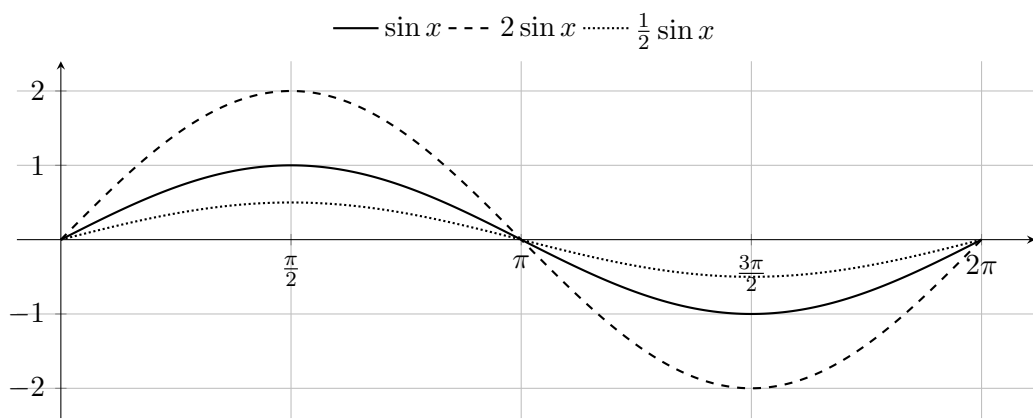
Label the points A-D with their coordinates and complete the table



| Angle            | $\cos \alpha$ | $\sin \alpha$ |
|------------------|---------------|---------------|
| 0                |               |               |
| $\frac{\pi}{2}$  |               |               |
| $\pi$            |               |               |
| $\frac{3\pi}{2}$ |               |               |
| $2\pi$           |               |               |

**Exercise 13: What does the factor in front do?**

Compare the three graphs.



a) Complete the table.

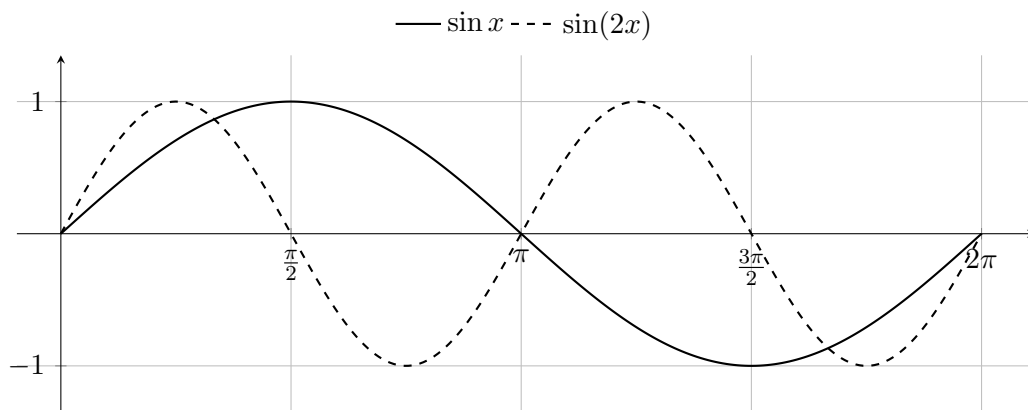
| Function             | Amplitude | Period |
|----------------------|-----------|--------|
| $\sin x$             |           |        |
| $2 \sin x$           |           |        |
| $\frac{1}{2} \sin x$ |           |        |

What changes when the factor in front of the sine changes?

## Part D continued: Frequency and Phase Shift

### Exercise 14: What does the factor inside do?

Compare  $y = \sin(x)$  and  $y = \sin(2x)$ .



a) How many complete oscillations does each graph make between 0 and  $2\pi$ ?

b) Which graph has the shorter period? \_\_\_\_\_

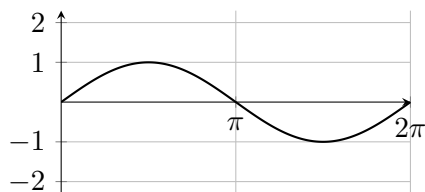
c) Complete: The period of  $\sin(2x)$  is \_\_\_\_\_.

d) Does the amplitude change? \_\_\_\_\_

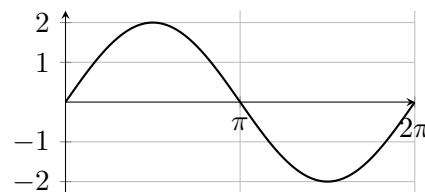
### Exercise 15: Match equations and graphs

Match each graph A–D to one equation. Write the correct letter next to each equation.

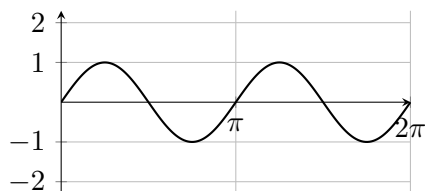
Graph A



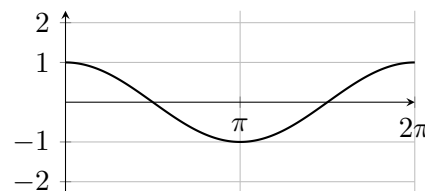
Graph B



Graph C



Graph D



• \_\_\_\_\_  $y = \sin(x)$

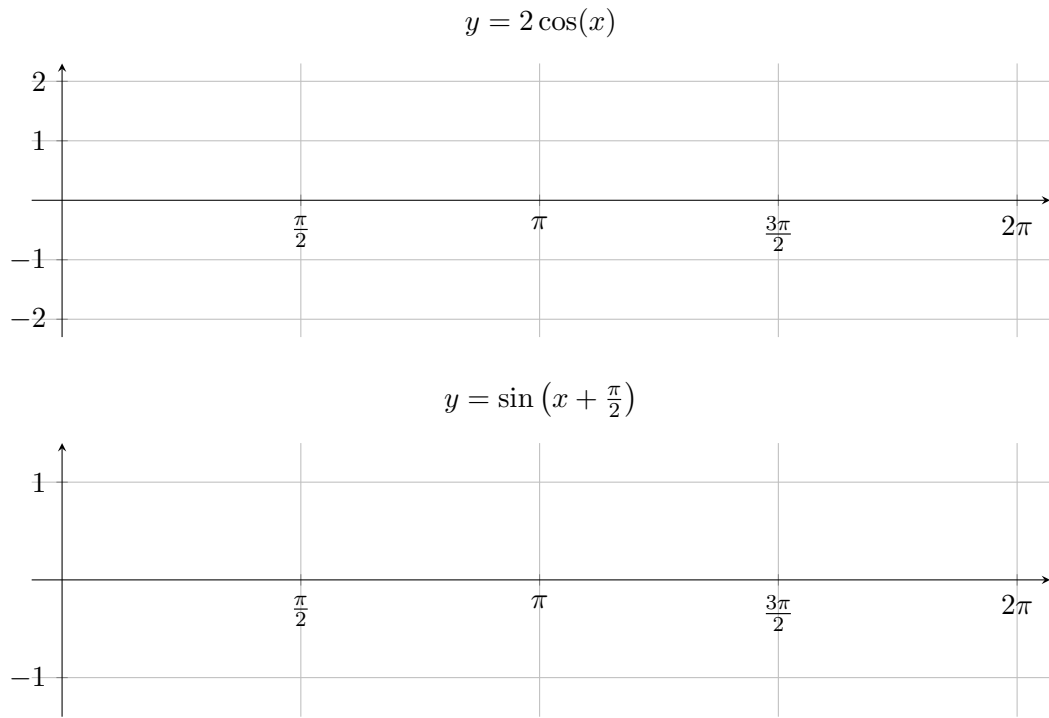
• \_\_\_\_\_  $y = 2\sin(x)$

• \_\_\_\_\_  $y = \sin(2x)$

• \_\_\_\_\_  $y = \cos(x)$

Part E: Sketching and Explaining

**Exercise 16: Sketch the graphs**  
Sketch one complete period of each function. Mark the zeros, maxima, and minima.



**Exercise 17: True or false?**  
Tick the correct box. Correct every false statement.

| Statement                                                         | True                     | False                    |
|-------------------------------------------------------------------|--------------------------|--------------------------|
| The sine function has period $2\pi$ .                             | <input type="checkbox"/> | <input type="checkbox"/> |
| The cosine function starts at 0.                                  | <input type="checkbox"/> | <input type="checkbox"/> |
| The maximum value of $3 \sin(x)$ is 3.                            | <input type="checkbox"/> | <input type="checkbox"/> |
| The function $\sin(2x)$ has amplitude 2.                          | <input type="checkbox"/> | <input type="checkbox"/> |
| Sine and cosine have the same shape but are shifted horizontally. | <input type="checkbox"/> | <input type="checkbox"/> |

Corrections:

---

**Challenge: Describe a transformed sine function**

Consider

$$f(x) = 3 \sin\left(\frac{x}{2} - \pi\right) + 1.$$

Without drawing the graph, determine:

Amplitude: \_\_\_\_\_ Period: \_\_\_\_\_

Horizontal shift: \_\_\_\_\_ Vertical shift: \_\_\_\_\_

Explain in one or two sentences how the graph differs from  $y = \sin(x)$ .

\_\_\_\_\_

*Remember: sine and cosine are not only triangle ratios. Their graphs are the basic mathematical shapes of oscillations and waves.*

### exercise 6:

Determine the period of the following functions and arrange them from the shortest period to the longest period

a)  $y = \sin(x) + 4$

d)  $y = \sin\left(\frac{1}{3}x\right)$

b)  $y = \sin\left(2x + \frac{\pi}{2}\right) + 1$

e)  $y = \cos(4x - 1) - 2$

c)  $y = \sin(3x)$

f)  $y = \cos\left(\frac{1}{4}x\right)$

### exercise 7:

Describe what happens in the following cases:

• the graph changes from  $y = \cos(x)$  to  $y = \cos(2x) + 2$

• the graph changes from  $y = \cos(x)$  to  $y = \cos(0,5x) - 3$

• the graph changes from  $y = \sin(x)$  to  $y = \sin(x - \pi)$

• the graph changes from  $y = \cos(x)$  to  $y = \cos\left(x + \frac{\pi}{2}\right)$

### exercise 8:

choose the correct function:

1. the graph has  
• amplitude 1  
• period  $\pi$

☐  $\sin(x)$       ☐  $\sin(0,5x)$

☐  $\sin(2x)$       ☐  $\sin(3x)$

2. the graph has  
• amplitude 1  
• period  $4\pi$

☐  $\cos(x)$       ☐  $\cos(0,5x)$

☐  $\cos(2x)$       ☐  $\cos(3x)$

3. which function is shifted to the right by  $\pi$  (compared to  $\sin(x)$ )

☐  $\sin(x + \pi)$       ☐  $\sin(2x)$

☐  $\sin(x - \pi)$       ☐  $\cos(x)$

4. Which function has the shortest period

☐  $\sin(x)$       ☐  $\sin(2x)$

☐  $\sin(5x)$       ☐  $\sin(0,25x)$

### exercise 9:

Complete the functions - fill in the missing numbers

1. the period should become  $\pi$ :

$y = \sin(\_\_x)$

2. the period should become  $4\pi$ :

$y = \sin(\_\_x)$

3. the period should become  $\frac{\pi}{2}$ :

$y = \sin(\_\_x)$

4. the graph shall be shifted to the right by  $\pi$ :

$y = \sin(x \pm \_\_)$

5. the graph should be shifted to the left by  $\frac{\pi}{2}$ :

$y = \cos(x \pm \_\_)$

## Modelling Real situations

### 1. two swinging pendulums

two pendulums swing with the same amplitude (1 cm)

- pendulum A completes one full swing every 4 seconds
- pendulum B completes one full swing every 2 seconds

both start at their highest position at  $t=0$

- write down a cosine function for each pendulum
- answer the following questions:

Which pendulum has the longer period?

Which pendulum moves faster?

Which graph is stretched more along the time axis?

Hint: it might be useful to sketch the graphs

### 2: Build your own oscillation:

Your oscillation should have:

- amplitude 1
- period  $\pi$
- shifted by  $\pi/2$  to the right

Write a possible sine function describing this wave.

Is there another possibility to express the function e.g. as a cos-function?